Improving the Localization Accuracy and Robustness of a UWB System Using VB-CSRUKF and RTS in Harsh Underground NLOS Environments

Bo Cao, Chunxia Jiang, Sumei Fan, Hua Zhang, and Wanli Liu

***Abstract*—Accurate and reliable location estimation of the shearer remains a crucial challenge to the realization of automated and unmanned mining. The poor positioning accuracy of conventional methods produces limited applicability owing to non-line-of-sight (NLOS) ranging errors, and the location errors of anchor nodes (ANs) and the velocity of the mobile target node (TN) are rarely considered. To tackle these issues, this article presents a new positioning strategy integrating calibration, the variational Bayesian (VB), constrained square root-unscented Kalman filter (CSRUKF), and robust Taylor series (RTS) algorithm to improve the ultra-wideband (UWB) positioning system accuracy in harsh underground coal mine environments. First, more accurate ANs position coordinates can be obtained using the proposed compensation method with the aid of reference nodes (RNs) under line-of-sight (LOS) conditions; then, the least squares (LSs) method is implemented to return the location of the TN using the compensated coordinates and corrected distance. Secondly, the VB-CSRUKF technique, which takes into account of the time-variant measurement noise and sigma points violating the constraints in the square root unscented Kalman filter (SRUKF), is used to eliminate the adverse effects of the NLOS error. Finally, the RTS method, in which robustness and the velocity of the TN in Taylor series are embedded, isadopted to further enhancethe final estimation accuracy. The experimental results corroborated that the developed VB-CSRUKF-RTS technique can efficiently and effectively ameliorate the overall accuracy, and is able to yield more accurate position estimation as compared with other approaches, possessing the high robustness and prominent performance.**

***Index Terms*—Constrained square root unscented Kalman filter (CSRUKF), robust Taylor series (RTS), ultrawideband (UWB), underground non-line-of-sight (NLOS) environment, variational Bayesian (VB).**

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Bo Cao is with the School of Mechanic Engineering, Anhui Science and Technology University, Chuzhou 233100, China, and also with the School of Mechanic Engineering, Hefei University of Technology, Hefei 230041, China (e-mail: shanxi2008caobo@163.com).

Chunxia Jiang, Sumei Fan, and Hua Zhang are with the School of Mechanic Engineering, Anhui Science and Technology University, Chuzhou 233100, China (e-mail: chunxia_njau@163.com; 1576160398@qq.com; chinafeihong@163.com).

Wanli Liu is with the College of Mechatronic Engineering, China University of Mining and Technology, Xuzhou 221116, China (e-mail: 4830@cumt.edu.cn).

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I. Introduction

AUTOMATED and intelligent mining is the cutting-edge technology being pursued for improving productivity and reducing major malignant accidents in the mining industry. The shearer is considered as one of the crucial pieces of mining equipment in a fully mechanized mining face (FMMF), the attitude and location information of which determine the corresponding state of the scraper conveyor and the hydraulic supports (HSs) during the production process [1], [2]. To realize mining automation, consequently, it is of great significance to determine the location of the shearer [3]. Although the inertial navigation system (INS) is available for the shearer positioning in the automated longwall mining, the localization performance deteriorates after the shearer completes a cutting cycle due to the inherent accumulated drift of the INS [4], which ultimately results in seriously inaccurate positioning. Hence, it is imperative to integrate with other localization techniques to calibrate errors induced by INS drift. The ultra-wideband (UWB) techniqueis considered a promising technologies in terms of both ranging and localization owing to its desirable properties, including low low-power consumption, centimeter-level range resolution, superior obstacle-penetration capacity, and immunization, to multipath interference [5]. This technique has been extensively exploited in the military, industrial production, logistics, and civilian fields, and it therefore has great application prospects for use in underground coal mines.

Several previous reports have investigated underground localization with the aid of a UWB system. In this research, a loosely- coupled [6], [7], or tightly- coupled INS/UWB [8], [9], [10] cooperative system was developed to improve the estimation accuracy. Wen et al. [8] developed a stable INS/UWB incorporated localization system for the shearer by utilizing the multi-model intelligent switching approach. Fan et al. [9] designed a sensor fusion approach via the use of UWB wireless radio and the INS based on

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a tightly- coupled model. Li et al. [10] established an error-state Kalman filter (KF) for harsh coal mine environments by integrating the results obtained from UWB system with INS measurements. The conventional positioning approaches are capable of providing sufficient positioning accuracy under line-of-sight (LOS) conditions, however, these strategies have substantial difficulty in yielding reliable and satisfactory estimation results in non-line-of-sight (NLOS) scenarios because the NLOS ranging error is always positive and much larger than the measurement noise, which restricts the applicability of these approaches.

Underground coal mine environments are usually extremely complicated and the direct paths between anchor nodes (ANs) and target nodes (TNs) are easy obstructed by dust, irregular objects, and moving target, which induces the reflection, scattering and diffraction, and ultimately larger NLOS errors [3], [11]. Thus, the NLOS error problem has been a popular research topic for UWB localization. The vast majority of related methods have been put forward to enhance the localization accuracy, and can be coarsely divided into two categories: 1) NLOS error identification methods and 2) NLOS error mitigation methods. The NLOS error identification techniques are mostly based on hypothesis testing [12], channel statistical analysis [13], support vector machine [14], and likelihood ratio test [15]. Regarding NLOS error mitigation, the main strategies focus on the weighted Taylor series method [16], semidefinite programming [17], interactive multi-model filter [18], KF [19], extended KF (EKF) [20], unscented KF (UKF) [21], particle filter [22] as well as cubaturcubature eKF (CKF) [23]. The KF provides the optimal solution for a linear system. The EKF implements only the first-order approximation to a nonlinear system, which results in a decrease in the accuracy of the state estimation and even divergence when the dynamic systems are strongly nonlinear and uncertain [24]. The PF usually has a high high-computational cost [25]. In contrast, the UKF using unscented transformation, does not require the calculation of Jacobian matrices calculation and its performance is better than that of the EKF [26], whereas the precision of the UKF is comparable to the performance of a third-degree CKF [27]. However, the aforementioned filters all have limits, and are incapable to ineffectively solve the smoothing problems since they require the definite statistics of noise in their calculations [28]; thus, they are impractical for obtaining certain measurement noise in most practical applications. The UWB signal propagation channel is more vulnerable to the surrounding environment, and the statistics of measurement noise may dynamically change with the varied position of the TN, especially in the complicated underground coal mine environments. To circumvent filtering accuracy deterioration, the variational Bayesian [29] (VB)-based filtering techniques have been successfully applied in state estimation problems to handle with the uncertain and changeable measurement noise. For example, the VB adaptive Kalman filter (VBAKF) with consideration of time-varying noise has been designed to diminish the influence of the NLOS error, and the better estimation accuracy can be achieved [30]. Moreover, by introducing an interacting multiple model (IMM), Cao et al. [31] embedded two VBAKFs in the IMM to diminish the LOS and NLOS errors, respectively, and the positioning accuracy was significantly improved. In [3], the VBUKF smoothing technique was used to eliminate the interference of the NLOS error, which showed that the NLOS ranging error was obviously diminished and the accuracy was considerably improved. Recently, the square-root UKF (SRUKF) was developed to enhance the filter performance of the UKF [32], and has been certified to have better effect than UKF due to its high high-estimation accuracy and easy implementation [33]. Moreover, to improve the robustness of SRUKF, the constrained SRUKF (CSRUKF) was developed to reduce NLOS error in which the sigma points were projected on to the constraint boundary to address the constrained optimization issues, and such constraints were found to effectively improve the state estimates, and better estimation accuracy than UKF was ultimately achieved [34]. Consequently, enlightened by this existing research, the VB approximation is embedded into the CSRUKF to overcome the filter performance degradation induced by uncertain and time-varying measurement noise, and ultimately to effectively alleviate the effect of the NLOS error.

Generally, in 3-dimensional (3D) UWB positioning system, the location of the TN is estimated by using the position coordinates of the ANs and the measured distances between the ANs and TN. It is generally known that the coordinates of the ANs should be installed and measured as accurately as possible before the UWB field tests are conducted. Unfortunately, in real-world applications, it is difficult to accurately measure the locations of ANs, especially in harsh mining environments, which degrades the localization accuracy. The chief influencing factors of the localization accuracy with respect to the UWB system are NLOS error and channel error (CE), of which the CE mainly includes the bias error (BE) along with scale-factor error (SFE) [3]. Since the SFE is systematic in nature, it could be compensated for based on appropriate calibration procedure. Various techniques have been developed and exploited to eliminate the BE to ameliorate the accuracy [35], [36], [37], [38]; however, these methods took no account of the effect of AN location errors on the position estimation of the TN. In many cases, it is impractical to accurately obtain the locations of ANs, and the ANs location errors have a great influence on the localization accuracy. Moreover, the aforementioned positioning methods rarely consider the location change of the moving TN within a period of observation, resulting in larger localization errors. This motivates us for the present research to simultaneously tackle both the AN location error and TN movement velocity in complicated environments with time-varying noise. Hence, to enrich the localization methodology and diminish inaccurate estimations induced by AN position errors and NLOS errors, a new approach that integrates calibration, VB approximation, the CSRUKF, and the robust Taylor series (RTS) algorithm is presented in this article, and the ANs location error, time-variant measurement noise, NLOS errors and TN movement speed are taken into consideration. More specifically, we utilize the calibration technique to compensate for the position errors of ANs, and the least-squares (LSs) approach is applied to estimate the position of the TN using AN's

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compensated position coordinates and the corrected distance. Subsequently, to account for the effect of the larger NLOS errors and unknown time-varying measurement noise, the CSRUKF fused-VB (VB-SRUKF) is performed to smooth the estimation results attained from the LS approach. Finally, the RTS method, considering robustness and the TN velocity, is applied to further improve the location accuracy. The main contributions of this research are summarized as follows:

1) The calibration scheme with the aid of reference nodes (RNs) is employed to reduce the effect of the AN location error on position estimation.
2) The VB-CSRUKF strategy based on a combination of the VB approach and CSRUKF is developed to cope with unknown measurement noise and sigma points violating the constraints in the SRUKF to eliminate the adverse effects of the NLOS error and ameliorate the positioning accuracy.
3) The RTS approach, which takes into consideration of the velocity of the TN and robustness, is introduced to further suppress the residual NLOS error, and extensive mobile experiments were carried out to demonstrate the rationality and superiority of the proposed VB-CSRUKF-RTS technique. The comparative experimental results validated the effectiveness and superiority of the developed approach, and the localization accuracy is verified to be significantly improved.

This article is organized as follows. The calibration method and VB-CSRUKF smoothing is described in Section II. The proposed approach is evaluated and performance comparisons are presented in Section III. The conclusions are given in Section IV.

II. Calibration Method and VB-CSRUKF Smoothing

A. Calibration Method

The range measurement represented by d_i in 3-D space between the TN and the ith AN can be modeled as follows:

$$d_i = (1+\beta_i) r_i + b_i + N_i + w_i \tag{1}$$

with

$$r_i = \sqrt{\left(x_i^A - x^m\right)^2 + \left(y_i^A - y^m\right)^2 + \left(z_i^A - z^m\right)^2} \tag{2}$$

where (x_i^A, y_i^A, z_i^A) and (x_i^m, y_i^m, z_i^m) are the positions of the i-th AN and TN, respectively, β_i indicates the SFE, b_{iI} is the BE; N_i is the NLOS ranging error, and w_i is white Gaussian noise.

Due to the unavoidable errors induced by the AN installation and measurement of the ANs, equation (1) is rewritten as:

$$d_i = (1+\beta_i)\sqrt{\left(x_i^m\right)^2 + \left(y_i^m\right)^2 + \left(z_i^m\right)^2} + b_i + N_i + w_i \tag{3}$$

where $\gamma_i^m = \widehat{\gamma}_i^A - \delta\gamma_i^A - \gamma^m$, $\gamma \in \{x, y, \text{ and } z\}$;$(\delta x_i^A, \delta y_i^A, \delta z_i^A)$ is the i-th AN location error, and $(\widehat{x}_i^A, \widehat{y}_i^A, \widehat{z}_i^A)$represents the erroneous location coordinates of the AN.

In the process of calibration, multiple appropriate RNs must to be chosen in LOS scenario and the collected measurement distance should only include the LOS ranging to effectively perform the calibration process. Additionally, the RNs location coordinates should be measured as accurately as possible. Thus, the measured distance d_{i-j} between and the jth RN and the i-th AN is expressed as follows:

$$d_{i-j} = (1+\beta_i)\sqrt{\left(x_{i-j}^R\right)^2 + \left(y_{i-j}^R\right)^2 + \left(z_{i-j}^R\right)^2} + b_i + w_{i-j} \tag{4}$$

where $\gamma_{i-j}^R = \widehat{\gamma}_i^A - \delta\gamma_i^A - \gamma_j^R$, $\gamma \in \{x, y, \text{ and } z\}$, and (x_j^R, y_j^R, z_j^R)denotes the coordinate of the j-th RN.

To effectively implement the calibration, one common RN is chosen among the RNs, and the remaining measurement differenced equations is formulated as follows:

$$\begin{aligned} d_{i-j} - d_{i-E} = {} & (1+\beta_i)\sqrt{\left(x_{i-j}^R\right)^2 + \left(y_{i-j}^R\right)^2 + \left(z_{i-j}^R\right)^2} \\ & -(1+\beta_i)\sqrt{\left(x_{i-E}^R\right)^2 + \left(y_{i-E}^R\right)^2 + \left(z_{i-E}^R\right)^2} + w_i \end{aligned} \tag{5}$$

where E indicates the common RN, and $i = 1, 2, \ldots, n-1$.

Then, equation (5) is extended using the first-order Taylor series at $(\widehat{x}_i^A, \widehat{y}_i^A, \widehat{z}_i^A)$, and the following matrix is formulated:

$$\Phi_i = H_i X_i + W \tag{6}$$

where $X_i = [\delta x_i^A, \delta y_i^A, \delta z_i^A, \alpha_i]^T$, and the definitions of$\Phi_i$ and H_i can be found in Appendix A.

The LS solution of the location error and the SFE concerning the i-th AN can be computed as follows:

$$\widehat{X}_i = \left(H_i{}^T H_i\right)^{-1} H_i{}^T \Phi_i. \tag{7}$$

Hence, via the use of the location error estimation, the location of each AN can be compensated as follows:

$$\begin{bmatrix} x_i^A \\ y_i^A \\ z_i^A \end{bmatrix} = \begin{bmatrix} \widehat{x}_i^A - \delta\widetilde{x}_i^A \\ \widehat{y}_i^A - \delta\widetilde{y}_i^A \\ \widehat{z}_i^A - \delta\widetilde{z}_i^A \end{bmatrix}. \tag{8}$$

To calculate the BE, the following matrix form is formulated:

$$U_i = M b_i + W \tag{9}$$

with

$$M = [1, \ 1, \ \ldots, \ 1]^T \tag{10}$$

$$U_i = \begin{bmatrix} d_{i-1} - \left(1+\widetilde{\beta}_i\right)\sqrt{\left(\widetilde{x}_{i-1}^R\right)^2 + \left(\widetilde{y}_{i-1}^R\right)^2 + \left(\widetilde{z}_{i-1}^R\right)^2} \\ d_{i-2} - \left(1+\widetilde{\beta}_i\right)\sqrt{\left(\widetilde{x}_{i-2}^R\right)^2 + \left(\widetilde{y}_{i-2}^R\right)^2 + \left(\widetilde{z}_{i-2}^R\right)^2} \\ \vdots \\ d_{i-n} - \left(1+\widetilde{\beta}_i\right)\sqrt{\left(\widetilde{x}_{i-n}^R\right)^2 + \left(\widetilde{y}_{i-n}^R\right)^2 + \left(\widetilde{z}_{i-n}^R\right)^2} \end{bmatrix} \tag{11}$$

where $\widetilde{\gamma}_{i-j}^R = \widehat{\gamma}_i^A - \widetilde{\gamma}_i^A - \gamma_j^R$, $\gamma \in \{x, y, \text{ and } z\}$.

Consequently, the BE of the i-th AN can be estimated as:

$$\widetilde{b}_i = \left(M^T M\right)^{-1} M^T U_i. \tag{12}$$

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B. Location Computation

Although the calibration approach is implemented to compensate for the position error of ANs, it is difficult to accurately estimate the TN's position coordinates due to the ranging errors in both LOS and NLOS scenarios. Based on a previous study [39], the ranging errors in the LOS condition are modeled as exponential distribution by conducting experiments with respect to UWB ranges under various environments and situations, which is applicable to any environment and expressed as:

$$f(d) = 0.1\left(1.01 - e^{-0.17d}\right) \tag{13}$$

where d represents the measurement distance. Thus, in this study, equation (13) is used to correct the measured distance from the UWB positioning system to obtain more accurate range estimates, which is given by the following:

$$\widehat{d} = d - f(d). \tag{14}$$

Based on the four corrected distances obtained from equation (14), the observation equation is converted into the matrix form, which is represented as follows:

$$\Psi X = S \tag{15}$$

where

$$S = \frac{1}{2}\begin{bmatrix} m_2 - m_1 + \widehat{d}_1^2 - \widehat{d}_2^2 \\ m_3 - m_2 + \widehat{d}_2^2 - \widehat{d}_3^2 \\ m_4 - m_3 + \widehat{d}_3^2 - \widehat{d}_4^2 \end{bmatrix}$$

$$\Psi = \begin{bmatrix} \left(x_2^A - x_1^A\right) & \left(y_2^A - y_1^A\right) & \left(z_2^A - z_1^A\right) \\ \left(x_3^A - x_2^A\right) & \left(y_3^A - y_2^A\right) & \left(z_3^A - z_2^A\right) \\ \left(x_4^A - x_3^A\right) & \left(y_4^A - y_3^A\right) & \left(z_4^A - z_3^A\right) \end{bmatrix} \quad X = [x,\ y,\ z]^T$$

and $m_i = (x_i^A)^2 + (y_i^A)^2 + (z_i^A)^2$, in which $\widehat{d}_1$, $\widehat{d}_2$, $\widehat{d}_3$, and $\widehat{d}_4$ indicate the corrected distance between the TN and four ANs, respectively; (x_i^A, y_i^A, z_i^A) is expressed as the i-th AN calibrated position with i=1, 2, 3, 4.

The solution of equation (15) using the LS method is given by:

$$X_{LS} = \left(\Psi^T \Psi\right)^{-1} \Psi^T S. \tag{16}$$

C. State Model

Via the use of the preceding equations, the initial position of the TN can be attained, however, the positioning accuracy will be insufficiently high due to the NLOS error. Thus, the VB-SRUKF-RTS technique is implemented to smooth the TN's estimation results obtained by the LS method. The state vector regarding the mobile TN at epoch $k$is represented as follows:

$$X_k = [p_k, v_k] = \left[x_k, y_k, z_k, \dot{x}_k, \dot{y}_k, \dot{z}_k\right]^T \tag{17}$$

where p_k=(x_k, y_k, z_k) and v_k=$(\dot{x}_k, \dot{y}_k, \dot{z}_k)$ indicate the location coordinates and velocity of the TN at epoch k, respectively.

Without loss of generality, the state-space model is given by:

$$X_k = BX_{k-1} + Cv_{k-1} \tag{18}$$

where $B = \begin{bmatrix} I_3 & \Delta t \cdot I_3 \\ 0 & I_3 \end{bmatrix}$, $C = \begin{bmatrix} (\Delta t^2/2) \cdot I_3 \\ \Delta t \cdot I_3 \end{bmatrix}$, I_3 is expressed as the 3-D identity matrix, Δt is sample period, and v_k indicates a Gaussian process with zero-mean vector and covariance Q_{k-1}.

The measurement vector in the LOS condition is given as:

$$Z_k = h(X_k) + n_k \tag{19}$$

with

$$h^i(X_k) = \left\|p_k - a^i\right\| = \sqrt{\left(x_k - x_i^A\right)^2 + \left(y_k - y_i^A\right)^2 + \left(z_k - z_i^A\right)^2} \tag{20}$$

where $a^i = (x_i^A, y_i^A, z_i^A)$ denotes the calibrated position coordinate of the i-th AN. The range between the TN and the i-th AN in LOS and NLOS scenarios can be represented as follows:

$$d_k^i = h^i(X_k) + n_k^i, \qquad i \in LOS \tag{21}$$

$$d_k^i = h^i(X_k) + \Upsilon_k^i + n_k^i, \qquad i \in NLOS \tag{22}$$

where n_k^i indicates the measurement noise, which is similar to the definition of v_k, and Υ_k^irepresents the NLOS error.

Since UWB signals are propagated through reflected, scattered, or diffracted paths in NLOS scenario, the estimated distance is greater than the true distance, expressed as follows:

$$\left\|p_k - a^i\right\| \le d_k^i. \tag{23}$$

To achieve higher positioning accuracy, we must remove the NLOS ranging from the measurement vector, leaving only the LOS measured distance, and the NLOS ranging considered as constraints is incorporated into the state-space model, which is expressed as follows:

$$\begin{cases} X_k = BX_{k-1} + Cv_k \\ Z_k = h(X_k) + n_k \\ \left\|p_k - a^i\right\| \le d_k^i. \end{cases} \tag{24}$$

D. Unconstrained VB-SRUKF

The SRUKF utilizes the square-root matrix of the error covariance to replace the covariance matrix of the conventional UKF in the process of iteration [40]. This can guarantee the positive definite of the covariance matrix and prevent divergence caused by accumulative error propagation in the UKF. The VB-SRUKF is the combination of the VB approach and SRUKF, of which the VB approximation is employed to compute the changeable process noise, and meanwhile the SRUKF is applied to update the system state estimation. The VB-SRUKF is implemented using the following procedure.

1) Time Update: Algorithmically, factorize the error covariance:

$$\Sigma_{k-1|k-1} = \Omega_{k-1|k-1}^T \Omega_{k-1|k-1}. \tag{25}$$

Evaluate the predicted state estimates and covariance matrix:

$$X_{k|k-1} = BX_{k-1|k-1} \tag{26}$$

$$\Sigma_{k|k-1} = B\Sigma_{k-1|k-1}B^T + CQC^T. \tag{27}$$

Equation (27) is reformulated as:

$$\Sigma_{k|k-1} = \left[B\Omega_{k-1|k-1}^T,\ CQ^{\frac{1}{2}}\right]\left[B\Omega_{k-1|k-1}^T,\ CQ^{\frac{1}{2}}\right]^T. \tag{28}$$

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Then, $\Omega_{k|k-1}$ can be obtained using the QR factorization, which is expressed as:

$$\Omega_{k|k-1} = qr\left\{\left[\Omega_{k-1|k-1}B^T,\ Q^{\frac{1}{2}}C^T\right]^T\right\}. \tag{29}$$

where the function $qr\{\cdot\}$represents QR factorization.

Estimate the measurement noise, given by [3]:

$$\begin{cases}\widehat{v}_{k|k-1} = \beta(\widehat{v}_{k-1|k-1} - n - 1) + n + 1\\ \widehat{V}_{k|k-1} = A\widehat{V}_{k-1|k-1}A^T.\end{cases} \tag{30}$$

2) Measurement Update: Compute the sigma points as following:

$$\begin{cases}F_{0,k|k-1} = X_{k|k-1}\\ F_{l,k|k-1} = X_{k|k-1} + \Omega^T_{k|k-1}\cdot\left(\sqrt{(n+\lambda)}\right)_l,\ \ l=1,\ldots,n\\ F_{l,k|k-1} = X_{k|k-1} - \Omega^T_{k|k-1}\cdot\left(\sqrt{(n+\lambda)}\right)_l,\ \ l=n+1,\ldots,2n\end{cases} \tag{31}$$

where the weight value w is declared as follows:

$$w_0^n = \frac{\lambda}{n+\lambda},\ w_0^c = \frac{\lambda}{n+\lambda} + 1 + \alpha^2 - \beta,\ w_l^n = w_l^c = \frac{\lambda}{2(n+\lambda)} \tag{32}$$

where $\lambda = \alpha^2(n+\delta) - n$ represents a composite scaling factor; in which δ is commonly set to 3-n, and the constant parameter α ($0 \leq \alpha \leq 0.5$) is used to determine the degree of dispersion of the sigma points. Via the utilization of the measurement model, the pseudo-measurement is computed as follows:

$$Z_{l,k|k-1} = h(F_{l,k|k-1}),\quad l = 0,\ldots,2n. \tag{33}$$

Calculate the predicted measurement value as follows:

$$\widehat{Z}_{k|k-1} = \sum_{l=1}^{2n} w_l^n Z_{l,k|k-1}. \tag{34}$$

Estimate the covariance matrix as follows:

$$\Sigma_{FZ,k|k-1} = \sum_{l=1}^{2n} w_l^c\left(F_{l,k|k-1} - X_{k|k-1}\right)\left(Z_{l,k|k-1} - \widehat{Z}_{k|k-1}\right)^T \tag{35}$$

$$\Sigma_{Z,k|k-1} = \sum_{l=1}^{2n} w_l^c\left(Z_{l,k|k-1} - \widehat{Z}_{k|k-1}\right)\left(Z_{l,k|k-1} - \widehat{Z}_{k|k-1}\right)^T. \tag{36}$$

Perform the following iteration (N steps for i =0, 1,..., N): Calculate the measurement noise variance matrix:

$$\widehat{R}_k^{(i)} = \left(\widehat{v}_{k|k} - n - 1\right)^{-1}\widehat{V}_k^{(i)}. \tag{37}$$

Calculate the innovation covariance:

$$\Sigma_{ZZ,k|k-1}^{(i+1)} = \Sigma_{Z,k|k-1} + \widehat{R}_k^{(i)}. \tag{38}$$

Calculate the upper triangular Cholesky factor Ω_Z of error covariance$\Sigma_{ZZ,k|k-1}$, which is expressed as:

$$\Omega_Z^{i+1} = qr\left\{\left[\zeta_Z^{(l)},\zeta_Z^{(l)},\ldots,\zeta_Z^{(2n)},\sqrt{\widehat{R}_k^{(i)}}\right]^T\right\} \tag{39}$$

where $\zeta_Z^{(l)} = \sqrt{w_l^c}(Z_{k|k-1}^{(l)} - \widehat{Z}_{k|k-1})$.

Compute the filter gain as following:

$$G_k^{i+1} = \Sigma_{SZ,k|k-1}\left(\Sigma_{ZZ,k|k-1}^{(i+1)}\right)^{-1} = \Sigma_{SZ,k|k-1}\left(\Omega_Z^{i+1}\right)^{-1}\left(\Omega_Z^{i+1}\right)^{-T}. \tag{40}$$

Update the state vector as following:

$$X_{k|k}^{i+1} = X_{k-1|k-1} + G_k^{i+1}\left(Z_k - \widehat{Z}_{k|k-1}\right). \tag{41}$$

Update error covariance matrix as following:

$$\Sigma_{k|k}^{i+1} = \Omega_{k|k-1}^T\Omega_{k|k-1} - G_k^{i+1}\left(\Omega_Z^{i+1}\right)^T\Omega_Z^{i+1}\left(G_k^{i+1}\right)^T. \tag{42}$$

Using equation (42), update the Cholesky factor of the estimated error covariance as the following representation:

$$\Omega_{k|k}^{i+1} = cholupdate\left\{\Omega_{k|k-1}, G_k^{i+1}\left(\Omega_Z^{i+1}\right)^T, -1\right\}. \tag{43}$$

Update measurement noise parameter as follows:

$$\widehat{V}_k^{(i+1)} = \widehat{V}_{k|k-1} + \sum_{l=1}^{2n} w_l^c\cdot\left(Z_k - h\left(X_{k|k}^i\right)\right)\left(Z_k - h\left(X_{k|k}^i\right)\right)^T. \tag{44}$$

Set $\widehat{V}_{k|k} = \widehat{V}_k^{(N)}$, $\Omega_{k|k} = \Omega_{k|k}^N$, $\widehat{R}_k = \widehat{R}_k^{(N)}$.

Herein, however, a more efficient approach is proposed to calculate $X_{k|k}^{i+1}$ and $U_{k|k}$, and the following is defined:

$$K_k^{i+1} = \Sigma_{SZ,k|k-1}\left(\Omega_Z^{i+1}\right)^{-1}. \tag{45}$$

The substitution of equation (46) into equation (40) yields:

$$G_k^{i+1} = K_k^{i+1}\left(\Omega_Z^{i+1}\right)^{-T}. \tag{46}$$

The state vector can be rewritten as follows:

$$X_{k|k}^{i+1} = X_{k|k-1} + K_k^{i+1}\left(\Omega_Z^{i+1}\right)^{-1}(Z_k - \widehat{Z}_{k|k-1}). \tag{47}$$

The substitution of equation (46) into equation (42) yields:

$$\Sigma_{k|k}^{i+1} = \Omega_{k|k-1}^T\Omega_{k|k-1} - K_k^{i+1}\left(K_k^{i+1}\right)^T. \tag{48}$$

Thus, the Cholesky factor of $\Sigma_{k|k}$ is updated as:

$$\Omega_{k|k}^{i+1} = cholupdate\left\{\Omega_{k|k-1}, K_k^{i+1}, -1\right\}. \tag{49}$$

E. Imposing the Inequality Constraints on the Estimates

Based on the aforementioned descriptions, the state vector along with error covariance is estimated via the VB-SRUKF, but some of the unconstrained sigma points that violates the inequality are not in the feasible region, thereby producing unreliable and inaccurate result estimation. To overcome this limitation, the constraints are incorporated in the SRUKF by projecting the unconstrained SRUKF sigma points onto the boundary of the feasible region at each time step to ameliorate the estimation performance of the SRUKF [40]. Thus, to successfully impose the inequality constraints of the NLOS distance on the state estimation $X_{k|k}$ and error covariance $U_{k|k}$, the new sigma points at epoch k yield the following:

$$\begin{cases}F_{k|k}^0 = X_{k|k}\\ F_{k|k}^{(l)} = X_{k|k} + \Omega_{k|k}^T\cdot\left(\sqrt{(n+\lambda)}\right)_l & l=1,\ldots,n\\ F_{k|k}^{(l)} = X_{k|k} - \Omega_{k|k}^T\cdot\left(\sqrt{(n+\lambda)}\right)_l & l=n+1,\ldots,2n.\end{cases} \tag{50}$$

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By considering the NLOS constraints, the constrained sigma point estimation problem is formulated as an optimization issue of the objective function to embody the projection process, expressed as follows:

$$\Psi\left(F_{k|k}^{(l)}\right) = \arg\min_{F}\left\{\left(F_{k|k}^{(l)} - \Re\right)^T W_k\left(S_{k|k}^{(l)} - \Re\right)\right\} \quad subject\ to \quad \left\|\Re(1:3) - a^i\right\| \le d_k^i, \quad i \in NLOS \tag{51}$$

where W_k indicates the symmetric positive definite weighting matrix with $W_k = \Sigma_{k|k}^{-1}$, which has been proven that utilizing $W_k = \Sigma_{k|k}^{-1}$ possesses the smallest estimation error covariance [41].

Based on equation (25), the objective function is rewritten as:

$$\Psi\left(F_{k|k}^{(l)}\right) = \arg\min_{F}\left\{(F_{k|k}^{(l)} - \Re)^T\Omega_{k|k}^{-1}\Omega_{k|k}^{-T}(F_{k|k}^{(l)} - \Re)\right\}. \tag{52}$$

To avoid the inverse calculation, the following is defined:

$$\chi = \Omega_{k|k}^{-T}\left(F_{k|k}^{(l)} - \Re\right). \tag{53}$$

Equation (53) can be formulated as follows:

$$\Re = F_{k|k}^{(l)} - \Omega_{k|k}^T u. \tag{54}$$

Based on a concept sourced from the existing literature [42], the optimization problem can be addressed more reliably, and the matrix $\Omega_{k|k}^T$ is partitioned as follows:

$$\Omega_{k|k}^T = \begin{bmatrix} \Xi_{11} & 0 \\ \Xi_{21} & \Xi_{22} \end{bmatrix} \tag{55}$$

where $\Xi_{11} \in \mathbb{R}^{3\times 3}$ and $\Xi_{22} \in \mathbb{R}^{(n-3)\times(n-3)}$.

Using equation (54) and (55),$\Re(1:3)$ can be obtained by:

$$\Re(1:3) = F_{k|k}^{(l)}(1:3) - \Xi_{11}\chi(1:3). \tag{56}$$

By applying equation (53) and (56), the objective function is reformulated as follows:

$$\min_{\chi}\left\{\chi^T(1:3)\ \chi(1:3) + \chi^T(4:n)\ \chi(4:n)\right\} \quad subject\ to \quad \left\|\Xi_{11}\chi(1:3) - \left(F_{k|k}^{(l)}(1:3) - a^i\right)\right\| \le d_k^i \quad i \in NLOS. \tag{57}$$

Since the constraints exist only for the first three moments of the state vector vector, without including $\chi(4:n)$, a reasonable choice is $\chi(4:n) = 0$, which can reduce the computational complexity. Thus, in this case, the optimization problem of equation (57) turns out to be:

$$\min_{\chi(1:3)}\left\{\chi^T(1:3)\ \chi(1:3)\ \right\} \quad subject\ to \quad \left\|\Xi_{11}\chi(1:3) - \left(F_{k|k}^{(l)}(1:3) - a^i\right)\right\| \le d_k^i \quad i \in NLOS. \tag{58}$$

Equation (58) is actually a convex optimization problem. By using the MATLAB extended optimization package YALMIP [43] and SeDuMi solver [44], the optimal $\chi(1:3)$ is obtained via the iterative technique [45], and the projected sigma points can then be calculated as follows:

$$\Psi\left(F_{k|k}^{(l)}\right) \triangleq \Re = F_{k|k}^{(l)} - \begin{bmatrix}\Xi_{11}\\ \Xi_{21}\end{bmatrix}\chi(1:3). \tag{59}$$

Using projected sigma points, calculate the following equations:

$$X_{k|k}^{\Psi} = \sum_{l=1}^{2n} w_l^n \Psi\left(F_{k|k}^{(l)}\right) \tag{60}$$

$$\Sigma_{k|k}^{\Psi} = \sum_{l=1}^{2n} w_l^c\left(\Psi\left(F_{k|k}^{(l)}\right) - X_{k|k}^{\Psi}\right)\left(\Psi\left(F_{k|k}^{(l)}\right) - X_{k|k}^{\Psi}\right)^T \tag{61}$$

Let $\zeta_{\Psi}^{(l)} = \sqrt{w^{(l)}}(\Psi(F_{k|k}^{(l)}) - X_{k|k}^{\Psi})$, the Cholesky factor of $\Sigma_{k|k}^{\Psi}$ is then calculated as follows:

$$\Omega_{k|k}^{\Psi} = qr\left\{\left[\zeta_{\Psi}^{(l)}, \zeta_{\Psi}^{(l)}, \ldots, \zeta_{\Psi}^{(2n)}\right]^T\right\}. \tag{62}$$

The updated state estimation and the error covariance matrix Cholesky factor at step k are, respectively, given by the following:

$$\begin{cases} X_{k|k} = X_{k|k}^{\Psi} \\ \Omega_{k|k} = \Omega_{k|k}^{\Psi}. \end{cases} \tag{63}$$

Thus, the position of the TN can be obtained as follows:

$$p_{k|k} = HX_{k|k}^{\Psi} \tag{64}$$

where $H = [1,\ 0,\ 0,\ 0,\ 0,\ 0;\ \ 0,\ 1,\ 0,\ 0,\ 0,\ 0;\ 0,\ 0,\ 1,\ 0,\ 0,\ 0]$.

The implementation of the VB-CSRUKF is presented in Table I.

F. Robust Taylor Series

On the basis of aforementioned calibration with respect to the location of each AN, the estimated location of the TN would be optimized by utilizing the VB-CSRUKF technique, however, it is difficult to yield an accurate position estimation when the NLOS ranges have larger NLOS errors. An effective iterative technique with higher localization accuracy is the robust Taylor series(RTS) algorithm that expands at an initial point and updates the TN position of the TN, and the final estimation is then approximated to the actual position. Accordingly, the RTS algorithm is implemented in the present study to further ameliorate the localization accuracy. To acquire more accurate location estimation with lower localization errors, the following equation is defined:

$$f_i(p) = \|p - a_i\| - d_i. \tag{65}$$

To enhance the robustness of the RTS technique, the M-estimators with consideration of the incorporation of an appropriate loss weight and the cost function are utilized for location estimation, which is formulated as follows:

$$p = \arg\min_{p}\sum_{i=1}^{m} s_i\rho(e) \tag{66}$$

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TABLE I
THE PROCESS OF THE VB-CSRUKF ALGORITHM

Initialization:

1. Set $X_{0|0}$, $\Sigma_{0|0}$, $\hat{v}_{0|0}$, $\hat{V}_{0|0}$.

Time update:

2. Factorize the error covariance using (25).
3. Calculate the predicted state estimates using (26), and calculate the covariance matrix using (27).
4. Calculate the measurement noise using (30).

Variational measurement update:

5. Set $X_{k|k}^{(0)}=X_{k|k}$, $\Sigma_{k|k}^{(0)}=\Sigma_{k|k}$, $\hat{v}_{k|k}=\hat{v}_{k|k-1}+1$, $\hat{V}_k^{(0)}=\hat{V}_{k|k}$.
6. Calculate the sigma points using (31).
7. Calculate the pseudo-measurement using (33).
8. Compute the predicted measurement using (34).
9. Estimate the cross-covariance matrix using (35).
10. Compute the innovation covariance without introducing the measurement noise using (36).
11. **For** $i=0:N-1$
12. $\hat{R}_k^{(i)}=(\hat{v}_{k|k}-n-1)^{-1}\hat{V}_k^{(i)}$
13. $\Sigma_{ZZ,k|k-1}^{(i+1)}=\Sigma_{Z,k|k-1}+\hat{R}_k^{(i)}$
14. $\Omega_Z^{i+1}=qr\left\{\left[\zeta_Z^{(l)},\zeta_Z^{(l)},\ldots,\zeta_Z^{(2n)},\sqrt{\hat{R}_k^{(i)}}\right]^T\right\}$
15. $K_k^{i+1}=\Sigma_{SZ,k|k-1}(\Omega_Z^{i+1})^{-1}$
16. $X_{k|k}^{i+1}=X_{k|k-1}+K_k^{i+1}(\Omega_Z^{i+1})^{-1}(Z_k-\hat{Z}_{k|k-1})$
17. $\Omega_{k|k}^{i+1}=cholupdate\left\{\Omega_{k|k-1},K_k^{i+1},-1\right\}$
18. $\hat{V}_k^{(i+1)}=\hat{V}_{k|k-1}+\sum_{l=1}^{2n}w_l^c\bullet(Z_k-h(X_{k|k}^i)(Z_k-h(X_{k|k}^i))^T$
19. **end for**
20. Set $\hat{V}_{k|k}=\hat{V}_k^{(N)}$, $\Omega_{k|k}=\Omega_{k|k}^{(N)}$, $\hat{R}_k=\hat{R}_k^{(N)}$.

Imposing the inequality constraints on the estimates

21. Calculate the sigma points using (50).
22. Solve the optimization problem of equation (58).
23. Calculate the projected sigma points using (59).
24. Calculate the constrained state estimate using (60).
25. Calculate the constrained Cholesky factor using (62).
26. Update the state estimate and Cholesky factor using (63)
27. Obtain the TN position using (64).
28. **Output**: $p_{k|k}$, $\hat{v}_{k|k}$, $\hat{V}_{k|k}$, $\hat{R}_{k|k}$.

where e, δp, and $\nabla f_i(p_0)$ are, respectively, given by:

$$e \approx f_i(p_0)+\nabla f_i(p_0)(p-p_0)=f_i(p_0)+\nabla f_i(p_0)\delta p \tag{67}$$

$$\delta p=\left[\delta_x,\delta_y,\delta_z\right]^T \tag{68}$$

$$\nabla f_i(p_0)=[a_i,\ b_i,\ c_i] \tag{69}$$

where $a_i=[(x_0-x_i^A)/(f_i(p_0))]$, $b_i=[(x_0-y_i^A)/(f_i(p_0))]$, $c_i=[(x_0-z_i^A)/(f_i(p_0))]$.

The optimal solution of equation (66) is determined by solving:

$$\sum_{i=1}^{m}\left[s_i\psi(e_i)\nabla f_i(p_0)^T\right]=0 \tag{70}$$

where $\psi(e_i)=\partial\rho/\partial e_i$ indicates influence function.

Let $\psi(e_i)/e_i=w_i$; equation (70) can then be rewritten as follows:

$$\sum_{i=1}^{m}\left[s_iw_ie_i\nabla f_i(p_0)^T\right]=0. \tag{71}$$

Equation (71) is written in the matrix form as follows:

$$MW\Theta=0 \tag{72}$$

where $\Theta=[f_1(p_0),f_2(p_0),f_3(p_0),f_4(p_0)]^T$, $W=\mathrm{diag}\{s_iw_i\}$ represents the equivalent weighted matrix; and

$$M=\begin{bmatrix}a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\\ a_4 & b_4 & c_4\end{bmatrix}. \tag{73}$$

Further, the following can be obtained by:

$$M^TWM\delta p-M^TW\Theta=0. \tag{74}$$

Thus, the solution can be obtained as follows:

$$\widehat{p}=p_0+(M^TWM)^{-1}M^TW\Theta. \tag{75}$$

Note that the value of w_i is called the equivalent weight function, which determines the robustness of an estimator, formulated as:

$$w_i=w(v_i)=\begin{cases}1 & |v_i|\le c\\ c/|v_i| & |v_i|>c\end{cases} \tag{76}$$

where $v_i=d_i-\|p-a_i\|$, and c is set within the interval [2.5,3].

The RTS algorithm possessed the capability of reducing the residual NLOS error and achieving more accurate position estimation. However, the RTS method taking no account of the velocity of the TN within a period of observation may still generate a large error, and the accuracy improvement of this method is not sufficient for some application scenarios with higher requirements in terms of the positioning accuracy. Thus, it is assumed that the mobile TN moves uniformly at the speed of v (v_x, v_y, v_z), and the position coordinates of the TN in a period of observation can be computed as follows:

$$\begin{cases}X_i=x+v_x\Delta t\\ Y_i=y+v_y\Delta t\\ Z_i=z+v_z\Delta t.\end{cases} \tag{77}$$

Due to the introduced velocity parameter, both the TN's position p (x, y, z) and the TN's velocity v (v_x, v_y, v_z) need to be estimated simultaneously. Hence, the virtual observation of the TN velocity parameter is introduced, expressed as follows:

$$\begin{cases}V_x=v_x+0\times v_y+0\times v_z\\ V_y=0\times v_x+v_y+0\times v_z\\ V_z=0\times v_x+0\times v_y+v_z.\end{cases} \tag{78}$$

By considering the speed of the TN, the TN velocity is embedded in the RTS approach to improve the estimated accuracy, and the final solution of the RTS can be obtained by:

$$\widetilde{p}=p_0+(\widetilde{M}^TW\widetilde{M})^{-1}\widetilde{M}^TW\widetilde{\Theta} \tag{79}$$

where $\widetilde{M}$ and $\widetilde{\Theta}$ is provided in Appendix B.

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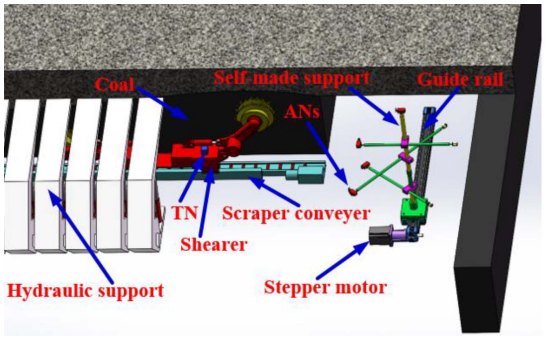


Fig. 1. UWB P440 sensors deployment in the roadway.

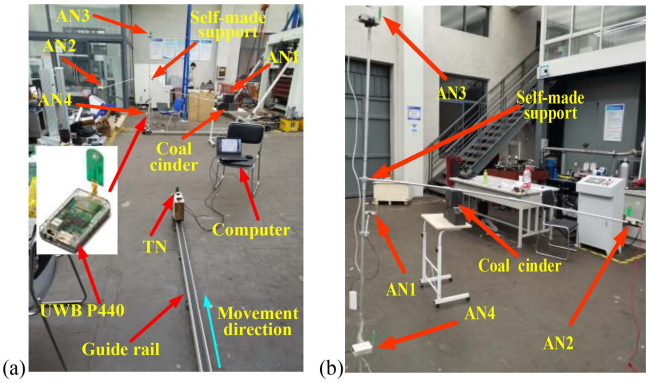


Fig. 2. (a) Experimental setup and (b) UWB arrangement scheme.

III. Experimental Results

A. Experimental Details

The UWB P440 module sensor was developed by the Time Domain Company [46], and more details about its predominant technical specifications are available ina previous publication [3]. The UWB P440 were selected as the ANs and TN for the experiment conducted in this research due to their remarkable superiority possessing high precision ranging. To obtain the shearer's position coordinates, it has been proposed that the ANs should be mounted on the HSs, and the TN be mounted on the shearer body [8], respectively. However, the main limitation of this technique is that it is difficult to accurately determine the positions of the ANs due to the fact that the HSs system moves forward frequently, which degrades the localization accuracy. To overcome this problem, it is proposed that the ANs of the UWB positioning system be mounted at the end of the FMMF to attain the location information of the shearer, as illustrated in Fig. 1. In the experiment, after the shearer completed a cutting cycle, the ANs are migrated a certain distance along the guide rail, and the new positions of ANs after migration can be calculated more accurately and quickly based on the last position of the ANs and the moved distance to make preparations for subsequent positioning. It is worth noting that this study only focused on UWB localization algorithm without considering the migration of the ANs along the guide rail to verify the suitability of the proposed algorithm in underground applications. The estimated accuracy of the UWB positioning system after the ANs migrates several times via the utilization of the proposed localization method will be investigated in future research.

To validate the suitability and effectiveness of the designed VB-CSRUKF-RTS technique, experimental campaigns were performed at the Center of Intelligent Mining Equipment of China University of Mining and Technology. To better simulate the shearer positioning and NLOS scenario utilizing the UWB system, the directed path of one AN was deliberately blocked using coal cinders to collect the NLOS ranging data. The four ANs were configured into a V-shaped pattern [3], [31]; this deployment configuration has been validated as preferable to achieve higher accuracy. The experimental layout of the UWB localization is shown in Fig. 2, in which the arrow pointed to the TN's movement direction. The TN was attached to the slider of the linear guide rail to simulate the shearer's movement, and the TN moved along the guide rail at respective velocities of 0.05, 0.075, 0.01, and 0.125m/s to take into account the effect of speed on the positioning accuracy. The four ANs were fixed on the self-made support with the respective locationsAN1 (1.895, 0.662, 0.875), AN2 (−0.473, 1.471, 1.172), AN3 (0.335, 0.102, 2.063), and AN4 (0.251, 0.034, 0.534).

TABLE II
Coordinates Positions of the ANs After Calibration Compensation

AN	SFE	BE	Coordinates
AN1	-0.0009	-0.1058	(1.911, 0.593, 0.904)
AN2	0.0046	-0.1485	(-0.503, 1.455, 1.192)
AN3	0.0041	-0.1042	(0.321, 0.094, 2.074)
AN4	-0.0083	0.0995	(0.268, 0.041, 0.513)

B. Experimental Results Evaluation

To highlight the goodness of the proposed approach based on the UWB positioning system, the position estimation error (e), along with absolute error involvement on the x-, y-, and z-axis, are defined as the main indexes by which to evaluate localization performance, and are expressed as following:

$$e = \sqrt{(x_e - x_a)^2 + (y_e - y_a)^2 + (z_e - z_a)^2} \quad (80)$$

$$\delta_x = |x_t - x_r|, \quad \delta_y = |y_t - y_r|, \quad \delta_z = |z_t - z_r| \quad (81)$$

where (x_a, y_a, z_a) is the actual position of the TN, and (x_e, y_e, z_e) is the estimated position of the TN obtained by using the involved localization algorithms in this study.

According to the aforementioned calibration technique, the SFE and BE for each AN were calculated, as reported in Table II, and the coordinate location of each AN could be effectively compensated by utilizing the location errors.

In order to demonstrate the suitability and robustness of the proposed technique, the state-of the-art RTS, UKF-RTS, SRUKF-RTS and CSRUKF-RTS schemes were respectively used to optimize the calculation results obtained from LS approach to comprehensively understand the merit of the VB-CSRUKF-RTS technique.

Fig. 3 exhibits the performance comparison concerning the TN location estimation error calculated by the mentioned positioning algorithms in the experimental evaluation scenario.

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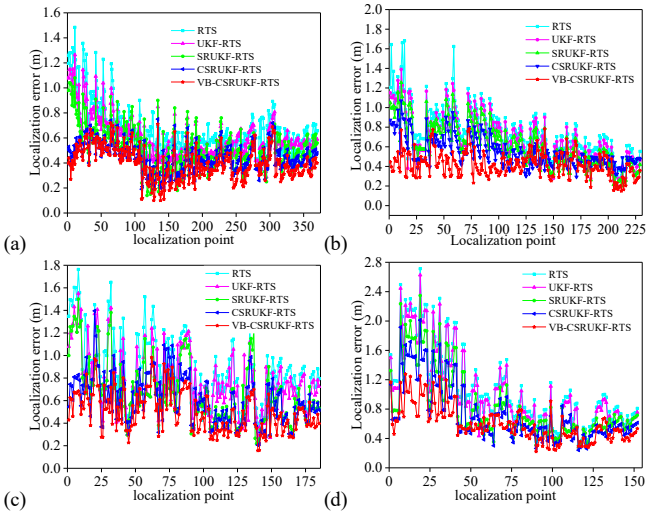


Fig. 3. Location estimation error achieved by utilizing the RTS, UKF-RTS, SRUKF-RTS, CSRUKF-RTS, and VB-CSRUKF-RTS approaches at different velocities: (a) 0.05 m/s; (b) 0.075 m/s;(c) 0.1 m/s; and (d) 0.125 m/s.

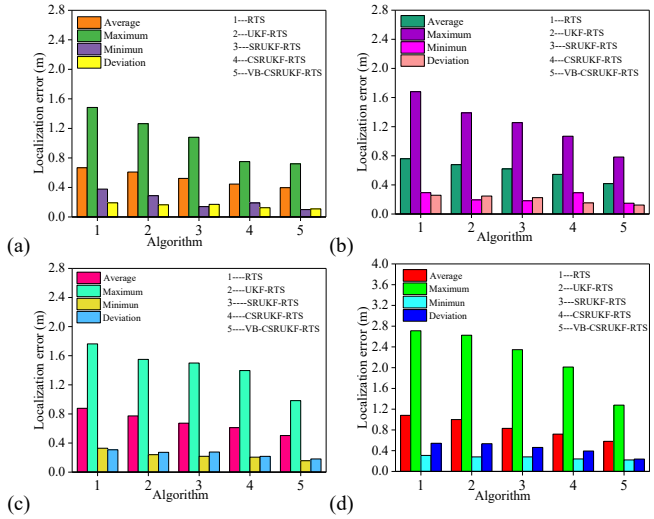


Fig. 4. Quantitative comparison of the statistical results of the localization error extracted from above approaches at different velocities: (a) 0.05 m/s; (b) 0.075 m/s; (c) 0.1 m/s; and (d) 0.125 m/s.

It is evident that the proposed VB-CSRUKF-RTS technique, possessing the lower positioning errors, substantially outperformed the other compared approaches, regardless of the velocity of the moving TN. In contrast, the RTS algorithm produced unacceptable and unreliable localization errors since it was significantly sensitive to the initial values. After the UKF was implemented for the results of the LS method, the NLOS errors were diminished and the accuracy was enhanced. Because the performance of the SRUKF was better than that of the UKF, the positioning accuracy of the SRUKF was higher. By incorporating the constraints with respect to the NLOS measurement distance into the SRUKF, the sigma points violating the constraints were projected onto the boundary of the feasible region at each step in the CSRUKF [41], which caused the CSRUKF method to achieve higher estimation accuracy than the SRUKF method. The VB method was embedded in the CSRUKF against the changeable measurement noise, which significantly improved the robustness and estimation accuracy of the CSRUKF, which consequently substantially outperformed the other methods. Additionally, the fluctuation ranges of the localization error increased as the velocity changed from 0.05 to 0.125 m/s for all the compared methods, meaning that a relatively accurate estimation can be achieved under the lower velocities.

To quantitatively investigate the positioning performance, the descriptive statistics regarding the maximum, minimum, average as well as standard deviation of the localization error extracted from the compared positioning approaches at different velocities are summarized in Fig. 4. We can see that the average localization accuracies of the proposed VB-CSRUKF-RTS technique at velocities of 0.05, 0.075, 0.1, and 0.125 m/s were the highest of all involved approaches, and were as high as 0.397, 0.417, 0.503, 0.58m, respectively; the performance of CSRUKF-RTS method was the second-best, with respective accuracies of 0.446, 0.545, 0.611 and 0.72 m. Similarly, the average accuracies of SRUKF-RTS method were, respectively, 0.523, 0.62, 0.673, and 0.83 m, those of the UKF-RTS method were, respectively, 0.609, 0.678, 0.773, and 1.00 m, and those of the RTS method were, respectively, 0.667, 0.759, 0.877, and 1.081m. Moreover, the VB-CSRUKF-RTS algorithm possessed the lowest the variance of the positioning error at four different velocities, indicating that more stable results were obtained as compared with the other approaches.

To further confirm the effectiveness and superiority of the designed methodology, the average localization accuracy improvement rate of each approach was calculated relative to the RTS technique, as illustrated in Fig. 5. It is evident from the figure that, via the utilization of the same positioning approach, the average accuracy improvement rate was significantly different at various speeds. The average localization accuracies achieved by VB-CSRUKF-RTS method corresponding to the velocity of 0.05, 0.075, 0.1, and 0.125 m/s were improved by 40.04%, 45.05%, 44.32%, and 46.34%, respectively. Similarly, the improvements of the UKF-RTS method were, respectively, 8.69%, 10.67%, 11.85%, and 7.49%, those of the SRUKF-RTS method were, respectively, 21.58%, 18.31%, 23.26%, and 23.42%, and those of the CSRUKF-RTS method were, respectively, 33.13%, 28.19%, 30.33%, and 33.39%. Obviously, the proposed VB-CSRUKF-RTS method possessing the higher accuracy improvement rates was able to robustly ameliorate the location accuracy of UWB system. The remarkable superiority of the VB-CSRUKF-RTS technique is chiefly attributable to the fact that the VB approximation was used to estimate the unknown measurement noise and sigma points that were outside the feasible region could be projected onto the boundary of the feasible region in the CSRUKF [41], and the RTS method considering the TN movement velocity, further suppressed the estimation error, which contributed to the achievement of better positioning accuracy as compared with the other strategies.

Table III illustrates the computational complexity of the mentioned approaches. Generally, the computational complexity of the method is closely related to its operational

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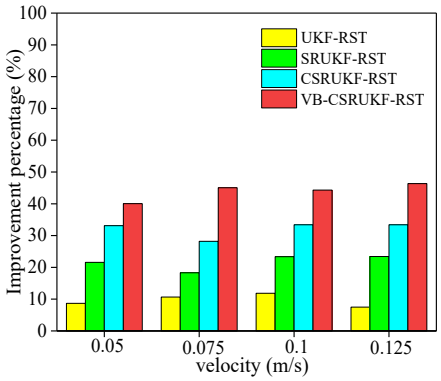


Fig. 5. Average localization accuracy enhancement relative to the RTS technique at different velocities.

TABLE III
COMPUTATIONAL COMPLEXITY OF DIFFERENT METHODS

Algorithm	Time (s)
RTS	0.066
UKF-RTS	0.145
SRUKF-RTS	0.188
CSRUKF-RTS	0.382
VB-CSRUKF-RTS	0.823

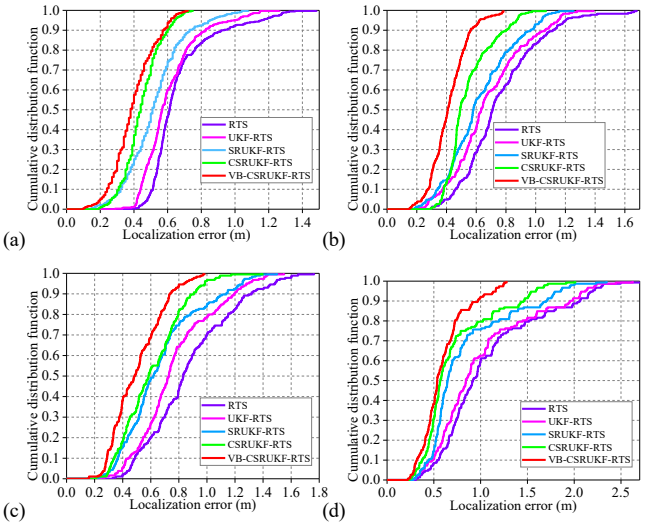


Fig. 6. The CDF of the localization error extracted from the RTS, UKR-RTS, SRUKF-RTS, CSRUKF-RTS, and VB-CSRUKF-RTS methods at different velocities: (a) 0.05 m/s; (b) 0.075 m/s; (c) 0.1 m/s; and (d) 0.125 m/s.

efficiency. Nevertheless, it is hard to retain a compromise balance between computational complexity and estimation accuracy. Obviously, the comparison reveals that the RTS approach has the lowest computational complexity among the five positioning methods, but this method generates the relatively large localization error. The VB methodology and CSRUKF algorithm require additional computation resources, which leads to the relatively large computational complexity of the proposed solution and is the most time consuming. Despite this, our method is capable of improving localization performance and yields a better positioning result when compared with the other methods.

Fig. 6 illustrates the cumulative distribution function (CDF) of the position estimation error extracted from the outcomes of the compared localization algorithms. Again, it is straightforward to see that the proposed VB-CSRUKF-RTS algorithm was superior to the compared approaches at different velocities. More specifically, the position estimation errors achieved byRTS, UKF-RTS, SRUKF-RTS, CSRUKF-RTS and VB-CSRUKF-RTS methods at the velocity of 0.05 m/s were approximately were 0.918, 0.826, 0.730, 0.601, and 0.570 m, respectively, as the CDF was up to 90%; those at the velocity of 0.075 m/s were 1.113, 1.061, 0.964, 0.799, and 0.563 m, respectively; those at the velocity of 0.1 m/s were 1.318, 1.196, 1.137,0.906, and 0.973 m, respectively; finally, those at the velocity of 0.125 m/s were 2.057, 1.954, 1.699, 1.457, and 0.882 m, respectively. These findings indicate the proposed VB-CSRUKF-RTS, which reflects the integration of the VB approach, constrained SRUKF and RTS method can efficiently diminish the position estimation error and robustly enhance the positioning performance, and, consequently, the proposed technique substantially outperformed the other four methods.

To deeply investigate the positioning performance of the developed VB-CSRUKF-RTS technique, Fig. 7 presents a detailed comparison of the estimated accuracy distributions on the three coordinate axes calculated by the compared approaches at the different velocities. The results intuitively reveal that the compared positioning methods achieved different estimation accuracies on the x-, y- and z-axes, and the localization errors were affected by the velocity of the TN. On all three axes for all velocities, the accuracy of the proposed VB-CSRUKF-RST technique substantially outperformed the other four methods. Although the localization errors of the three axes could be effectively diminished after applying VB approach and CSRUKF, the positioning errors still increased with the increase of the velocity. At the velocity of 0.05 m/s, the proposed method that integrates the VB approach, CSRUKF, and RTS algorithm reduced the average localization errors of the x-, y-, and z-axes from 0.332 m, 0.404 m, and 0.306 m to 0.204 m, 0.187 m, and 0.135 m, respectively, and the corresponded average accuracies were enhanced by 35.54%, 53.71%, and 55.88%, respectively.

Similarly, at the velocity of 0.075 m/s, the average location estimation accuracies were enhanced by 46.09%, 51.38%, and 12.5%, respectively; those at the velocity of 0.1 m/s were enhanced by 48.07%, 58.52%, and 48.68%, and those at the velocity of 0.125 m/s was improved by 58.16%, 4.73%, and 6.25%, respectively.

These improvements demonstrated that the proposed scheme was significantly more effective and robust than the other compared methods. Furthermore, the reduction in the fluctuation range concerning the localization errors on the three axes demonstrates that the proposed VB-CSRUKF-RTS strategy was superior to the other approaches.

In summary, according to the results of the comparative analysis, it can be concluded that the proposed VB-CSRUKF-RST technique is capable of achieving effective improvement for UWB system in terms of localization accuracy under different velocities of the moving TN, and was found to possess high robustness and feasibility.

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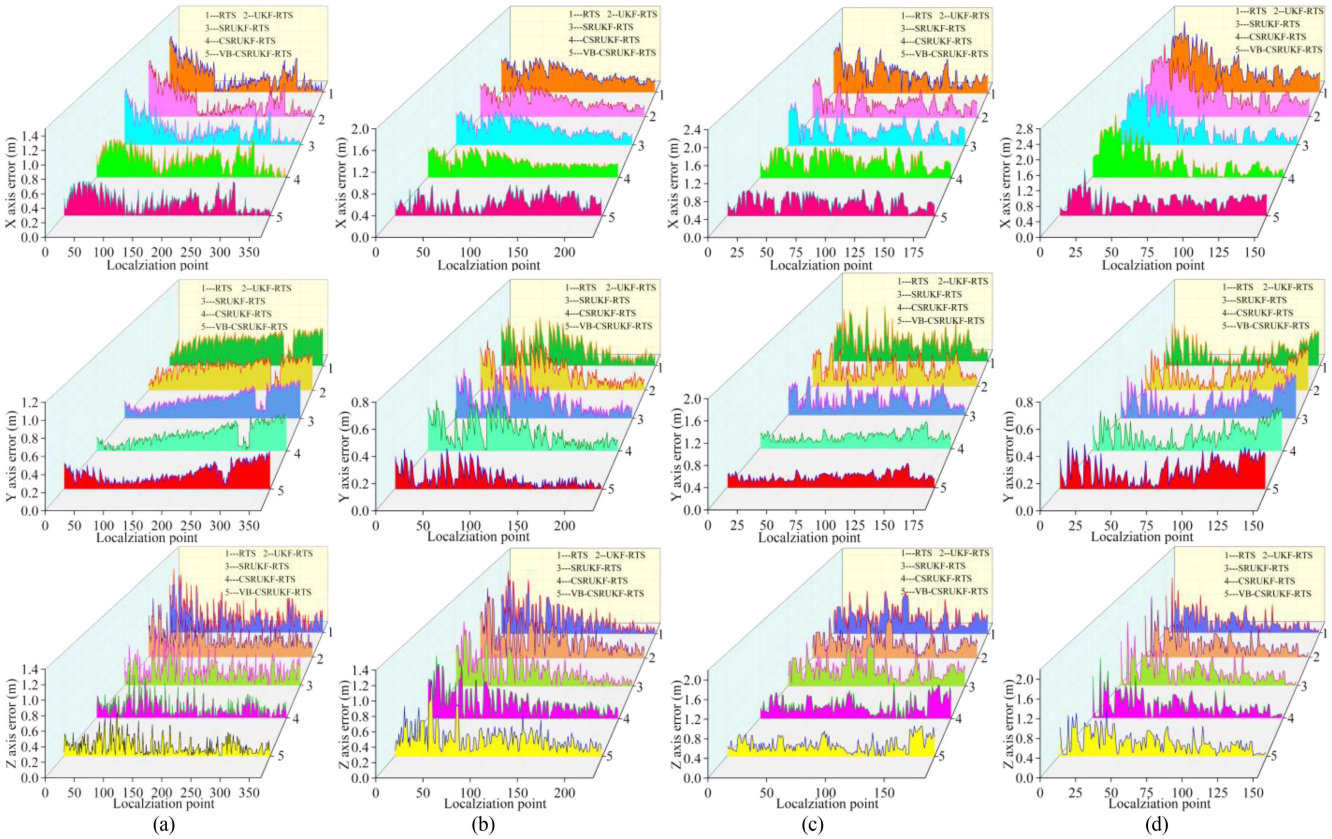


Fig. 7. The comparison Comparison of the positioning error distributions on the three axes achieved by RTS, UKR-RTS, SRUKF-RTS, CSRUKF-RTS, and VB-CSRUKF-RTS approaches at different velocities. From top to bottom, the localization error distributions on the x-, y- and z-axes are, respectively, presented. (a) 0.05 m/s. (b) 0.075 m/s. (c) 0.1 m/s. (d) 0.125 m/s.

IV. Conclusion

To diminish the effect of the AN location error and NLOS ranging errors, this article proposes a novel technique that incorporates the VB, CSRUKF and RTS methods based on calibration to improve the UWB positioning system accuracy for harsh underground coal mines environments. The SFE, BE, and more accurate ANs position coordinates can be obtained by applying the proposed calibration method. In the proposed method, the LS approach is adopted to calculate the location of the TN using the compensated coordinates and corrected distance. Subsequently, the VB-CSRUKF strategy based on the VB approach and CSRUKF is implemented to handle with time-variant measurement noise and sigma points violating the constraints in the SRUKF to reduce the adverse effects of NLOS errors. Then, the RTS approach taking into account of both the robustness and the velocity of the TN is implemented to further suppress the residual NLOS error to optimize the estimated results. Experiments were carried out under different TN velocities to highlight the effectiveness and superiority of applying the developed VB-CSRUKF-RTS technique to UWB system. The comparative experimental results demonstrate that the proposed methodology can effectively ameliorate the overall average estimation accuracy, and that the higher accuracy can be achieved under the lower velocities. Compared with using only RTS method, the VB-CSRUKF-RTS method can introduce overall accuracy improvement rates of up to 40.04%, 45.05%, 44.32%, and 46.34%, corresponding to the TN velocities of 0.05, 0.075, 0.1, and 0.125 m/s, respectively, and is found to possess the stronger robustness and better localization performance. Note that in UWB system, the limitations of the developed algorithm possesses the longest running time, and we will optimize the estimation performance of the algorithm for reducing the complexity. Additionally, future work will focus on and the validation of the developed technique in real underground applications and the investigation of the positioning accuracy after the ANs migrated several times.

Appendix A

Equation (5) is extended using the first order Taylor series at $(\widehat{x}_i^A, \widehat{y}_i^A, \widehat{z}_i^A)$, which is expressed as:

$$\begin{aligned} d_{i-j} - d_{i-E} &\cong (1+\beta_i)\left(\widehat{d}_{i-j} - \frac{\widehat{x}_i^A - x_j^R}{\widehat{d}_{i-j}}\delta x_i^{AN} - \frac{\widehat{y}_i^A - y_j^R}{\widehat{d}_{i-j}}\delta y_i^A - \frac{\widehat{z}_i^A - z_j^R}{\widehat{d}_{i-j}}\delta z_i^A\right) \\ &-(1+\beta_i)\left(\widehat{d}_{i-E} - \frac{\widehat{x}_i^A - x_c^R}{\widehat{d}_{i-E}}\delta x_i^A - \frac{\widehat{y}_i^A - y_c^R}{\widehat{d}_{i-E}}\delta y_i^{AN} - \frac{\widehat{z}_i^A - z_c^R}{\widehat{d}_{i-E}}\delta z_i^A\right) + w_i \\ &\cong \widehat{d}_{i-j} - \widehat{d}_{i-E} + (\widehat{d}_{i-j} - \widehat{d}_{i-E})\beta_i - A_{i-j}^x \delta x_i^A - A_{i-j}^y \delta y_i^A - A_{i-j}^z \delta z_i^A + w_i \end{aligned} \tag{82}$$

where $A_{i-j}^{\gamma} = ([(\widehat{\gamma}_i^A - \gamma_j^R)/(\widehat{d}_{i-j})] - [(\widehat{\gamma}_i^A - \gamma_E^R)/(\widehat{d}_{i-E})])$, $\gamma \in \{x, y, \text{ and } z\}$.

Then, equation (A1) is formulated as the following matrix:

$$\Phi_i = H_i X_i + W \tag{83}$$

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where

$$\Phi_i = \begin{bmatrix} (d_{i-1} - d_{i-E}) - \left(\widehat{d}_{i-1} - \widehat{d}_{i-E}\right) \\ (d_{i-2} - d_{i-E}) - \left(\widehat{d}_{i-2} - \widehat{d}_{i-E}\right) \\ \vdots \\ \left(d_{i-(n-1)} - d_{i-E}\right) - \left(\widehat{d}_{i-(n-1)} - \widehat{d}_{i-E}\right) \end{bmatrix} \tag{84}$$

$$H_i = \begin{bmatrix} -A^x_{i-1} & -A^y_{i-1} & -A^z_{i-1} & \left(\widehat{d}_{i-1} - \widehat{d}_{i-E}\right) \\ -A^x_{i-2} & -A^y_{i-2} & -M^z_{i-2} & \left(\widehat{d}_{i-2} - \widehat{d}_{i-E}\right) \\ \vdots & \vdots & \vdots & \vdots \\ -A^x_{i-(n-1)} & -A^y_{i-(n-1)} & -A^z_{i-(n-1)} & \left(\widehat{d}_{i-(n-1)} - \widehat{d}_{i-E}\right) \end{bmatrix}. \tag{85}$$

APPENDIX B

By considering the velocity of the TN, $\nabla f_i(p_0)$ and δp are replaced with $\nabla \widetilde{f}_i(p_0)$ and $\widetilde{\delta} p$, respectively, expressed as follows:

$$\nabla \widetilde{f}_i(p_0) = [a_i,\ b_i,\ c_i,\ a_i\Delta t,\ b_i\Delta t,\ c_i\Delta t,\ ] \tag{86}$$

$$\widetilde{\delta} p = \left[\delta_x, \delta_y, \delta_z, v_x, v_y, v_z\right]^T. \tag{87}$$

Then, $\widetilde{\Theta}$ and $\widetilde{M}$ in equation (81) are, respectively, expressed as:

$$\widetilde{\Theta} = \left[f_1(p_0), f_2(p_0), f_3(p_0), f_4(p_0), V_x, V, V_z\right]^T \tag{88}$$

$$\widetilde{M} = \begin{bmatrix} M & \Delta t \times M \\ 0 & I_3 \end{bmatrix} \tag{89}$$

where I_3 is the 3-D identity matrix.

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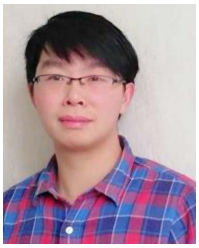
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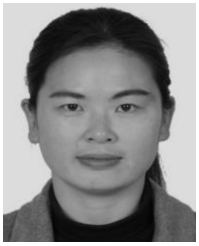
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Bo Cao received the Ph.D. degree in mechanical design and theory from the China University of Mining and Technology, Xuzhou, China, in 2022.

He is currently a Lecturer with Anhui Science and Technology University, Chuzhou, China. He is currently a Postdoctoral Fellow with Hefei University of Technology, Hefei, China. His current research interests include wireless sensor network, improvement positioning accuracy, underground localization and navigation, and intelligent perception principle of mechanical equipment.



Chunxia Jiang received the Ph.D. degree in mechanical engineering from the Nanjing Forestry University, Nanjing, China, in 2019.

She is currently a Lecturer with the Anhui Science and Technology University, Chuzhou, China. Her current research interests include intelligent perception principle of mechanical equipment, and intelligent equipment development.



Sumei Fan received the undergraduate degree in mechanical engineering from the Anhui Science and Technology University, Chuzhou, China, in 2023, where she is currently pursuing the master's degree.

Her main research interests is indoor navigation and positioning.



Hua Zhang received the Ph.D. degree in mechanical engineering from Nanjing Forestry University, Nanjing, China, in 2016.

He is currently a Professor with Anhui Science and Technology University, Chuzhou, China. His current research interests include intelligent perception principle of mechanical equipment, intelligent equipment development, and agricultural equipment localization.



Wanli Liu received the Ph.D. degree from Tianjin University, Tianjin, China, in 2016.

He is an Associate Professor with Jiangsu Province and Education Ministry Co-Sponsored Collaborative Innovation Center of Intelligent Mining Equipment, China University of Mining and Technology, Xuzhou, China. His research interests include terrestrial laser scanning, remote sensing, unmanned and intelligent vehicle, and underground position and navigation.

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